ADVANCED OPTICAL MATERIALS

Supporting Information

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Ultra-Compact and NonVolatile Nanophotonic Neural Networks

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Supplemental Material

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Supplementary Note 1

Refractive index of different materials

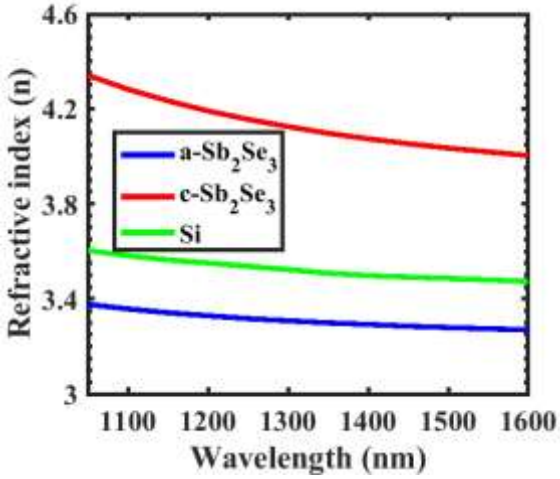

Figure S1. (a) Refractive indices n of a-Sb_2Se_3, c-Sb_2Se_3, and silicon from 1050 to 1600 nm [1].

Supplementary Note 2

Manufacturing tolerance analysis

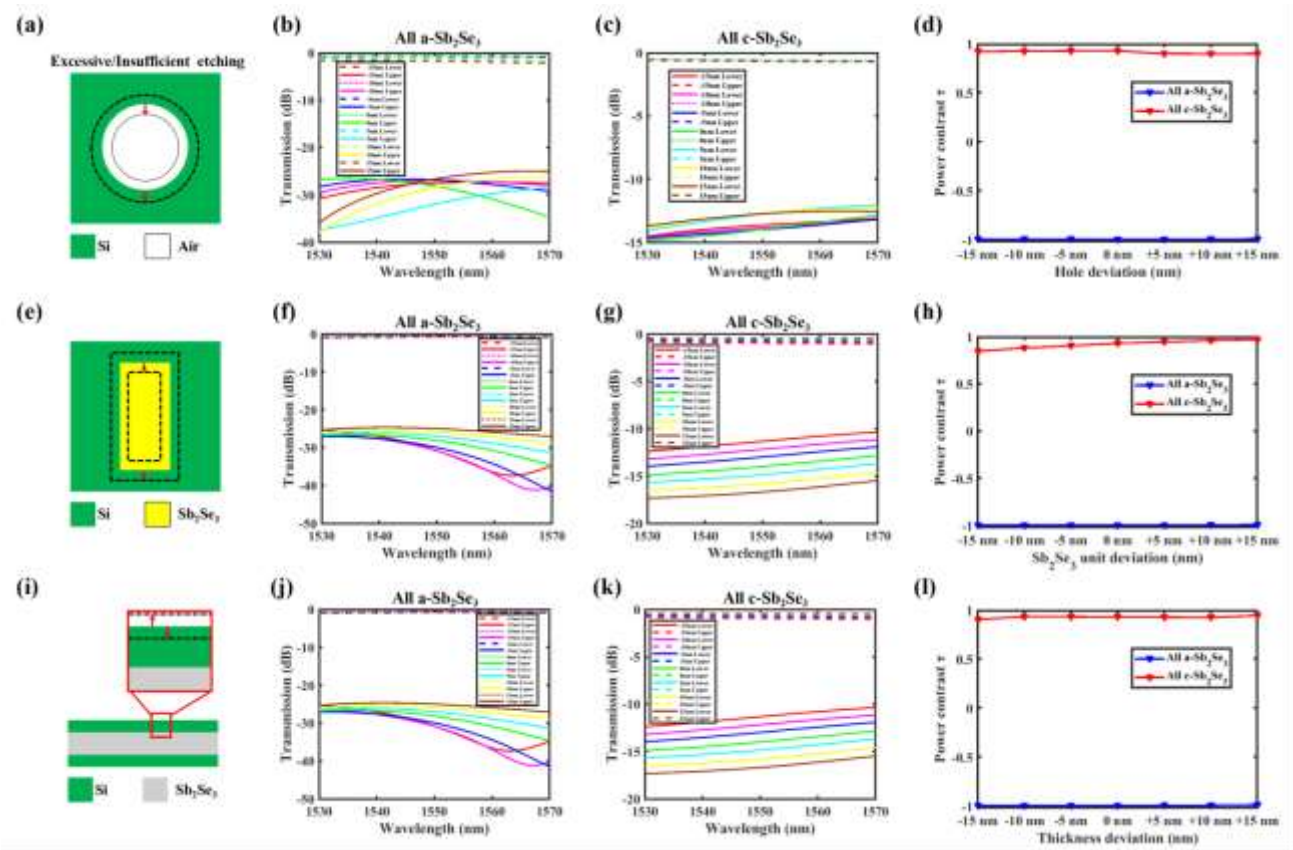

Figure S2. Analysis of manufacturing tolerance for the 1×2 nanophotonic scattering unit. (a) Circular hole deviation structure model. (b, c) Simulated the transmission spectrum as the hole diameters varies from -15 nm to +15 nm. (d) Simulated the power contrast (at 1550 nm) as the hole diameters varies from -15 nm to +15 nm. (e) Sb_2Se_3 unit deviation structure model. (f, g) Simulated the transmission spectrum as the Sb_2Se_3 units size (Length and width) varies from -15 nm to +15 nm. (h) Simulated the power contrast (at 1550 nm) as the Sb_2Se_3 units size varies from -15 nm to +15 nm. (i) Top layer silicon deviation structure model. (j, k) Simulated the transmission spectrum as the top silicon thickness varies from -15 nm to +15 nm. (l) Simulated the power contrast (at 1550 nm) as the top silicon thickness varies from -15 nm to +15 nm.

Expansion (over-etching) or contraction (under-etching) of pattern profiles are typical defects in device fabrication. As mentioned earlier, the structure was designed on a silicon layer with a thickness of 220 nm on the top of the SOI substrate. To investigate the manufacturing tolerances of nanophotonic units, we simulate the effects of varying the top hole radius, Sb_2Se_3 length and width, and silicon thickness on device performance. Moreover, taking 1550nm wavelength as an example, we also analyze the influence of structure size change on power contrast τ.

Its transmission spectrum is given in Fig. S2, and certain manufacturing tolerance obtained. Fig. S2a, e, and i respectively show the structural model of changing the diameter of the round hole, the size of the Sb_2Se_3 element and the thickness of the top layer silicon. Fig. S2b,c,f,g,j,k respectively show the transmission performance analysis of the corresponding structure (circular hole, Sb_2Se_3 element, silicon waveguide) size variation. Fig. S2d, h and l respectively show the variation of power contrast at 1550 nm wavelength with the structural size deviation. From the deviation degree of the curve, it can be seen that when the structure size changes within a certain range (-15 nm ~ +15 nm), the transmission efficiency deviation of the output channel is small, and the manufacturing deviation of the device has strong robustness.

Supplementary Note 3

Classification and recognition results of neural networks with four different resolutions

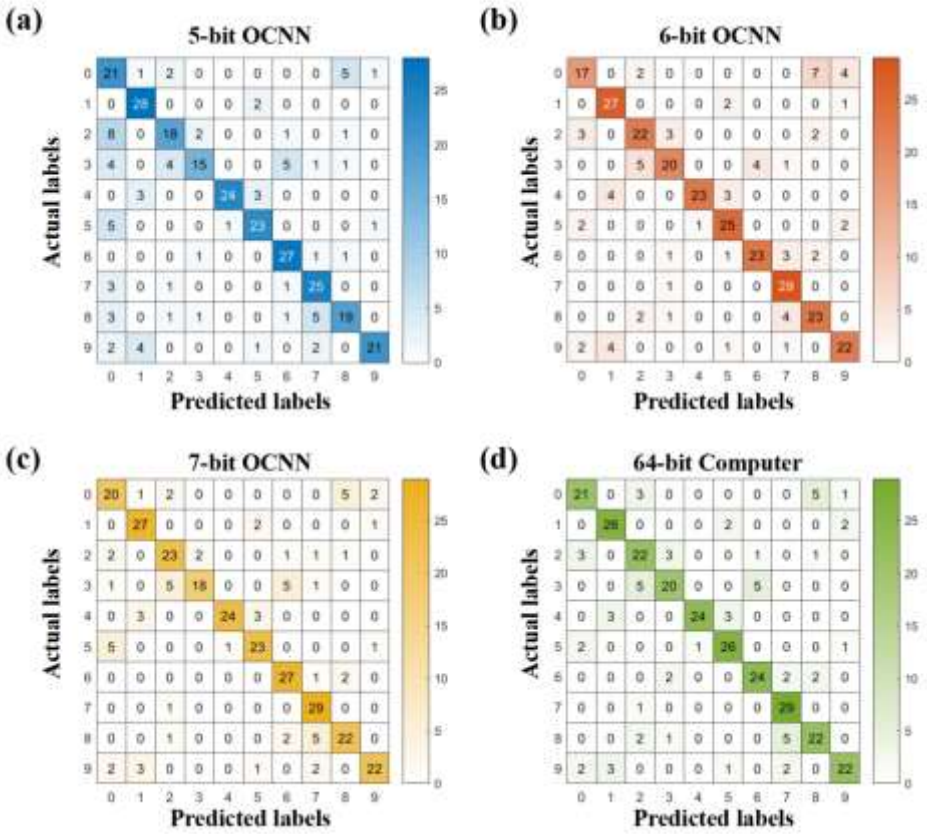

Figure S3. Speech sequence classification and recognition results of three different PNN simulations and speech sequence classification results of 64-bit computers.

Supplementary Note 4

Inverse design algorithm optimization process

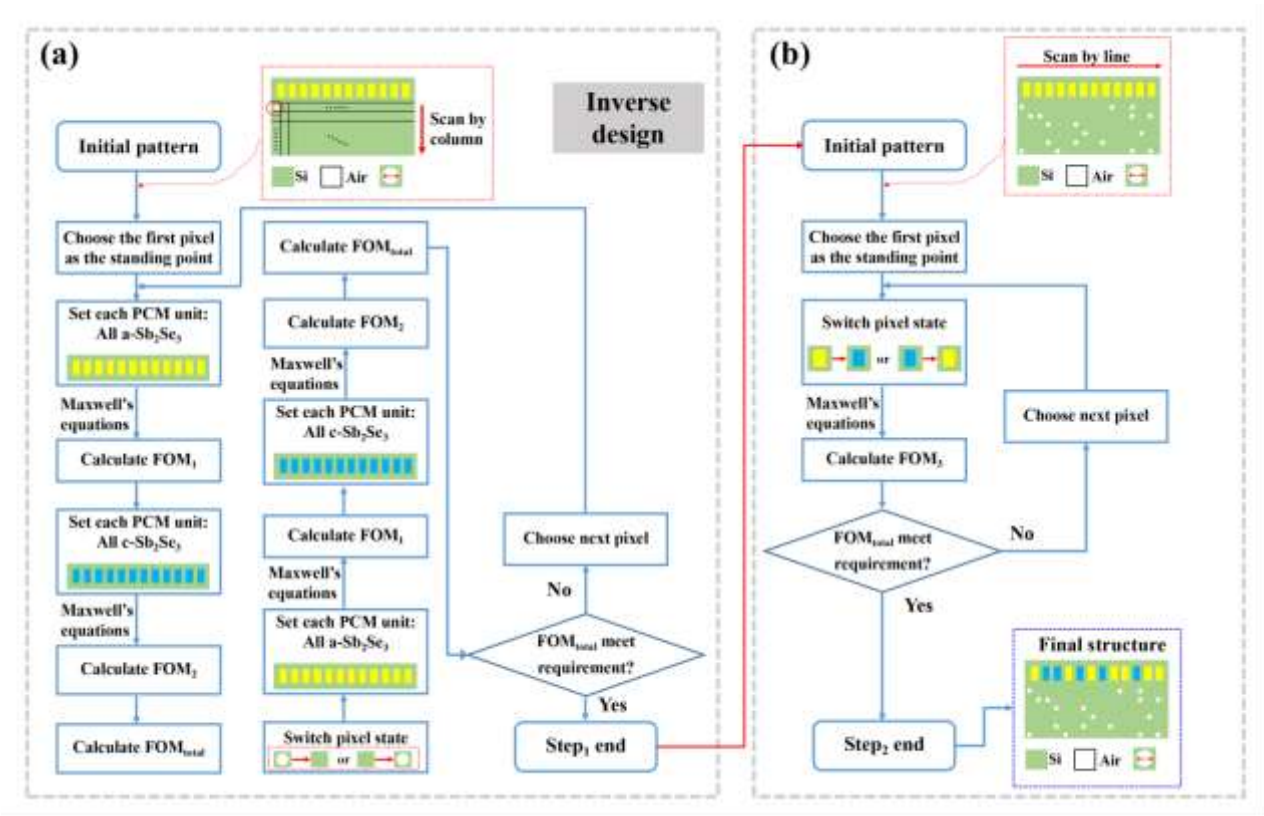

Figure S4. The general process of inverse design of optical nano-micro structure (take this nanophotonic scattering unit as an example).

To obtain the computational structure of photon vector matrix multiplication with 7-bit distinguishable power contrast characteristics, it is necessary to introduce inverse design [2-6]. As one of the key algorithms, direct binary search (DBS) algorithm has attracted more and more attention in recent years. As shown in Fig. S4, we take the nanophotonic scattering unit as an example to introduce the optimization process of inverse design. Three-dimensional finite Difference Time domain (3D FDTD) method was used for simulation. In the optimized region, the pixels have binary dielectric properties: Si/air or a-Sb_2Se_3/c-Sb_2Se_3 (corresponding to the logical state "0" or "1", respectively). We divided the coupling region into two parts: an air hole array and a Sb_2Se_3 array. Basically, the optimization process consists of two steps.

Step 1 (Optimize the distribution of air holes) :

First, we assume that when the Sb_2Se_3 array is in an amorphous state, the incident light passes through the coupling region and transmits through the lower output channel. Otherwise, it is transmitted through the upper output channel. Select the first air hole pixel as the starting point. We switch this pixel state and calculate the figure of merit (FOM) that represents the device's performance. If FOM is improved, this pixel state is retained. If not, it returns to its previous state.

The DBS algorithm then proceeds to select the next pixel for computation. An iteration ends when all pixels (12 × 40) have been counted. Iterations continue until there is no further improvement in FOM or the device's performance remains stable. The FOM of the initial structure is defined as:

$$FOM_1 = T_{lower} - T_{upper}$$

$$FOM_2 = T_{upper} - T_{lower}$$

$$FOM_{total} = \frac{FOM_1 + FOM_2}{2}$$

where T_{upper} and T_{lower} are the average transmission efficiency of the upper and lower output channels, respectively. The transmittance data are discrete at different wavelengths; therefore, we chose 101 data points distributed in the wavelength range of 1530-1570 nm at equal distances.

Step 2 (Optimize the distribution of Sb_2Se_3 array) :

The tuning process is the same as the first step. Select the first Sb_2Se_3 cell pixel as the starting point. We switch this pixel state and calculate the FOM representing the device's performance. A total of 128 discrete linear distribution target power contrasts $\tau_i \in$ (-0.9995,0.9274) are defined. The FOM of the step 2 is defined as:

$$FOM_3 = -abs\left(\frac{T_{upper} - T_{lower}}{T_{upper} + T_{lower}} - \tau_i\right), i = 1,2,\cdots,128.$$

Figure S5a, b respectively show the iterative process of the above two steps. Fig. S5c shows the difference between the 128 distinguishable contrasts and the ideal contrast obtained by the inverse design algorithm.

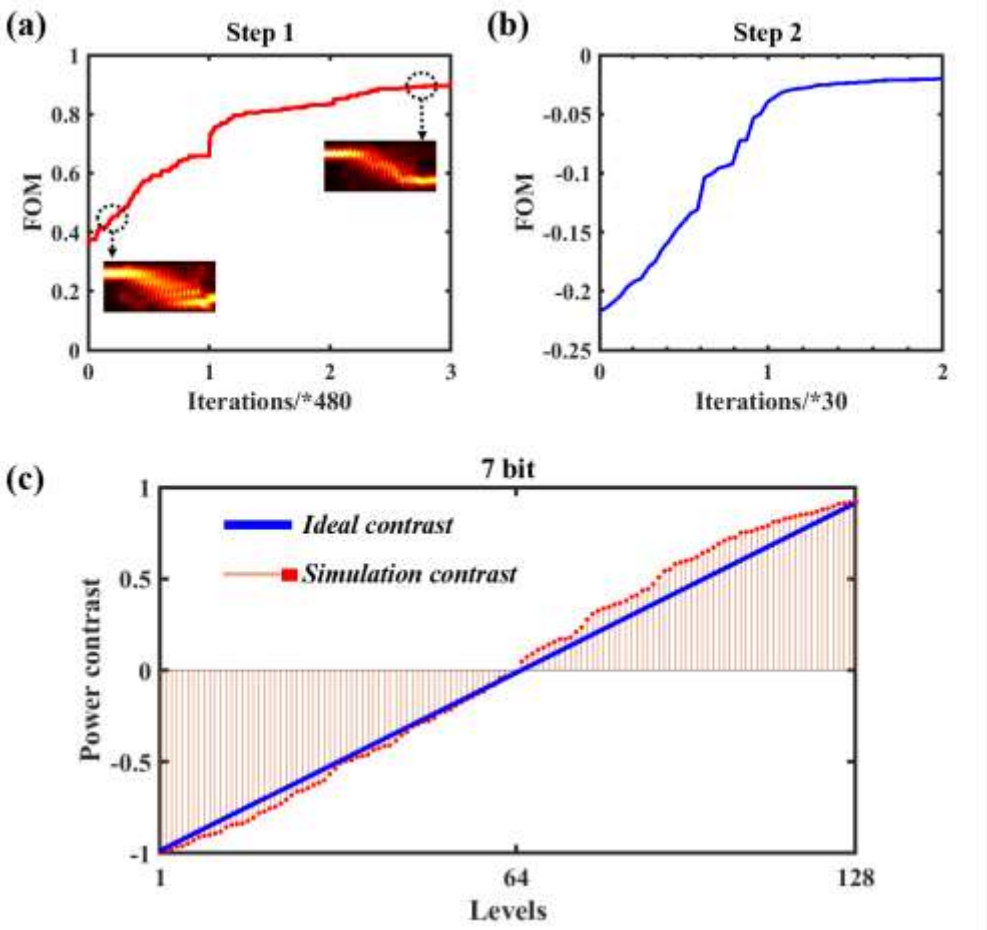

Figure S5. (a) Air round hole array optimization iteration process, each iteration contains 480 simulations. (b) Sb_2Se_3 array optimization iteration process, calculated 128 different contrast average FOM, each iteration contains 30 simulations. (c) The ideal contrast and the contrast results simulated by the nanophotonic scattering unit.

Supplementary Note 5

Nanophotonic scattering unit scheme with high precision resolution

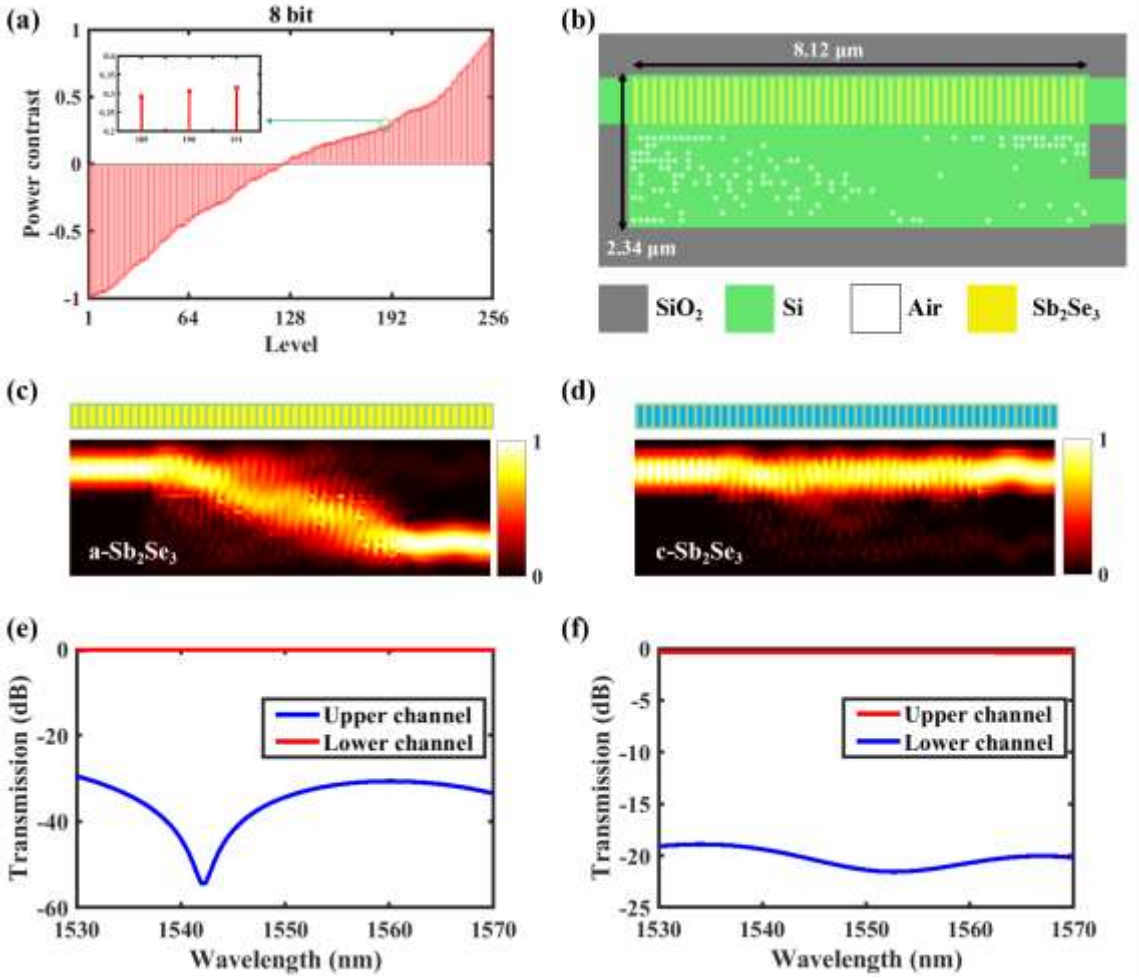

Figure S6. (a) The contrast results simulated by the nanophotonic scattering unit. (b) High resolution 1×2 programmable nanophotonic units. (c, d) The energy density at 1550 nm in both crystalline and amorphous conditions was simulated using the FDTD. We plotted the electromagnetic energy density $U= \epsilon|E|^2 + \mu|H|^2$. (e, f) The spectral transmission curves of nanophotonic units in the upper and lower channels are simulated.

After the basic building blocks of a PNN with a 7-bit resolution, we then extend the nanophoton unit to higher resolution. As shown in Fig. 6b, we changed the length of the coupling region to 8.12 μm and increased the number of Sb_2Se_3 arrays to 50. Based on the nanophotonic structure, we obtain Fig. S6a show the 256 (8 bit) distinguishable level by coding and regulating the state of Sb_2Se_3 array. Fig. S6c shows the electromagnetic energy densities corresponding to a-Sb_2Se_3 and c-Sb_2Se_3 at λ = 1550 nm. Fig. S6e and Fig. S6f show the transmission curves of the two output channels in the wavelength range of 1530-1570 nm. At 1550 nm, the insertion losses (ILs) of the device are −0.214 dB (a-Sb_2Se_3: lower channel) and −0.321 dB (c-Sb_2Se_3: upper channel), and the crosstalk values are −34.358 dB (a-Sb_2Se_3: upper channel) and −21.381 dB (c-Sb_2Se_3: lower channel). In the 40 nm bandwidth range, the average transmission efficiency between different output channels is higher than 94%, and the average crosstalk is lower than 0.9%.

Next, MNIST and Fashion-MNIST datasets were used to test network classification recognition.

The classification results of the N-PNN with 8-bit resolution are shown in Fig. S7a, b. It can be seen that the classification accuracy reaches 88.03% (Fashion-MNIST) and 98.74% (MNIST), respectively. These classification accuracy is almost the same as that of 64-bit computers.

On the other hand, we also find that the distortion of power contrast (8bit case) is more severe than that of 7bit case. This indicates that higher levels (such as 9bit) classification accuracy will be difficult. The possible reasons for limitations may come from the air hole deviation disturbance, the limited Sb_2Se_3 units, etc. Fortunately, the classification accuracy is already very close to the 64 bit computer level, which means that higher levels (such as 9bit) is not particularly urgent from a practical perspective.

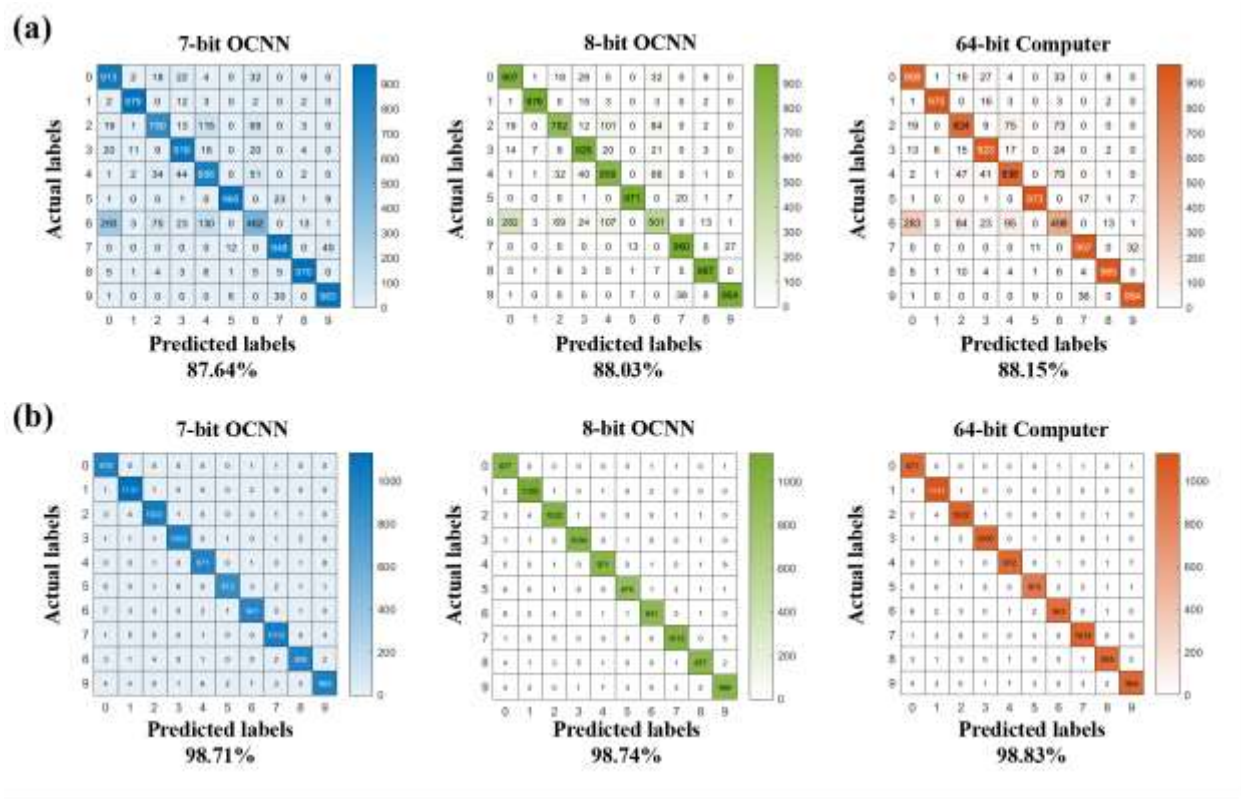

Figure S7. (a) Fashion-MNIST datasets classification and recognition results of two different PNN simulations and the results of 64-bit computers. (b) MNIST datasets classification and recognition results of two different PNN simulations and the results of 64-bit computers.

Supplementary Note 6

Comparison of compute density

Refs.	[7]	[8]	[9]	This work

MAC Size	9 × 4/8 × 8	32 × 32	2 × 2	**32 × 32**
MAC Area (mm^2)	0.285×0.354/ 0.127×0.127	~2	~0.131	**~0.1**
TOPS	0.4/0.5	164	0.2	**164**
Compute Density ($TOPS/mm^2$)	1.2/162	82	1.04	**1640**
Clock Speed(GHz)	12/18	10	17	**10**
Weight precision	5 bit	6 bit	9 bit	**7 bit**

Table S1. Comparison of compute density of various integrated photonic computing hardware.

To further compare the performance of different photon computing schemes, we introduce computing density as an index. The photonic core compute density can be defined as [7-9]:

$$Compute\ Density = \frac{Tera\ Operations\ Per\ Sencond(TOPS)}{Chip\ Area(mm^2)}$$

For the single core unit we proposed, its footprint is only 4.92 × 2.34 $\mu m^2 \approx 1.15\times10^{-5}$ mm^2. Taking into account the incidental footprint area (Such as input and output waveguide) required for MAC operation, we estimated the total footprint area of each nanophotonic unit (The size is only 15 × 5 μm^2), and a 32 × 32 array would have an area of less than 0.1 mm^2. We assume that the array size is 32 × 32 and the data rate is 10 GHz. The compute density of the network is about 1640 $TOPS/mm^2$. Compared with other work, this optical network composed of compact nanophotonic structures enables a huge increase in compute density.

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